

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Zinc, reference standard solution 1000 ppm

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>SZ13-100, SZ13-500</b>                                                                            |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product contains one or more substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Substances/mixtures corrosive to metal Category 1

#### Health hazards

Skin Corrosion/Irritation Category 1 B  
Serious Eye Damage/Eye Irritation Category 1

#### Environmental hazards

No hazards identified

#### Label Elements



Corrosion

Signal Word

Danger

Hazard Statements

H290 - May be corrosive to metals

H314 - Causes severe skin burns and eye damage

Precautionary Statements

P234 - Keep only in original packaging

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P310 - Immediately call a POISON CENTER or doctor

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P363 - Wash contaminated clothing before reuse

P390 - Absorb spillage to prevent material damage

P402 - Store in a dry place

P406 - Store in corrosion resistant polypropylene container with a resistant inliner

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

Other information

## Section 3 - Composition and Information on Ingredients

| Component                   | CAS No    | Weight % |
|-----------------------------|-----------|----------|
| Water                       | 7732-18-5 | 93.38    |
| Nitric acid ...% [C ≤ 70 %] | 7697-37-2 | 6.5      |
| Zinc oxide                  | 1314-13-2 | 0.12     |

## Section 4 - First Aid Measures

Inhalation

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

Ingestion

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

|                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                           |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. . Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                         |

## Section 5 - Fire Fighting Measures

### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Hazardous Decomposition Products**

Nitrogen oxides (NO<sub>x</sub>).

### **Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### **Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### **Environmental Precautions**

Should not be released into the environment.

### **Methods for Containment and Clean Up**

#### **Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

#### **Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### **Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Do not store in metal containers.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]  
Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component                   | Australia                                                                             | New Zealand WEL                                                                                                    | ACGIH TLV                                              | The United Kingdom                                       | Germany                                                                                                                                                        |
|-----------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] | STEL: 4 ppm<br>STEL: 10 mg/m <sup>3</sup><br>TWA: 2 ppm<br>TWA: 5.2 mg/m <sup>3</sup> | TWA: 2 ppm<br>TWA: 5.2 mg/m <sup>3</sup><br>STEL: 4 ppm<br>STEL: 10 mg/m <sup>3</sup>                              | TWA: 2 ppm<br>STEL: 4 ppm                              | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min | TWA: 1 ppm (8 Stunden). AGW -<br>TWA: 2.6 mg/m <sup>3</sup> (8 Stunden). AGW -                                                                                 |
| Zinc oxide                  | STEL: 10 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>   | TWA: 0.1 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup><br>STEL: 0.5 mg/m <sup>3</sup><br>STEL: 5 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> |                                                          | TWA: 0.1 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.4 mg/m <sup>3</sup><br>Höhepunkt: 4 mg/m <sup>3</sup> |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls**

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

**Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Butyl rubber   | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommended Filter type:</b>        | and maintenance of respiratory protective devices<br>Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                         |
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                                                                                                                |
| <b>Hygiene Measures</b>                | Keep away from food, drink and animal feeding stuffs. When using do not eat, drink or smoke. Contaminated work clothing should not be allowed out of the workplace. Provide regular cleaning of equipment, work area and clothing. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wear suitable gloves and eye/face protection. |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                                                                                                          |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                |                                          |
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Odor</b>                                    | Odorless                 |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | < 1                      | Acidic                                   |
| <b>Melting Point/Range</b>                     | 0 °C / 32 °F             |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 100 °C / 212 °F          |                                          |
| <b>Flash Point</b>                             | Not applicable           | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | > 1 (ether = 1)          |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | 14 mmHg                  |                                          |
| <b>Vapor Density</b>                           | 0.7                      | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.1                      |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                        | No information available |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Nitric acid ...% [C ≤ 70 %]                    | -2.3                     |                                          |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Explosive Properties</b>                    | No information available |                                          |
| <b>Oxidizing Properties</b>                    | No information available |                                          |

### Other information

## Section 10 - Stability and Reactivity

|                               |                                            |
|-------------------------------|--------------------------------------------|
| <b>Reactivity</b>             | None known, based on information available |
| <b>Stability</b>              | Stable under normal conditions.            |
| <b>Conditions to Avoid</b>    | Incompatible products, Excess heat.        |
| <b>Incompatible Materials</b> | Metals, Reducing Agent, Strong bases.      |

**Hazardous Decomposition Products** Nitrogen oxides (NOx).

**Hazardous Polymerization** Hazardous polymerization does not occur.

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

(a) acute toxicity;  
**Oral** Based on available data, the classification criteria are not met  
**Dermal** No data available  
**Inhalation** No data available

#### Toxicology data for the components

| Component                   | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation           |
|-----------------------------|---------------------------|------------------------------|---------------------------|
| Water                       | -                         | -                            | -                         |
| Nitric acid ...% [C ≤ 70 %] |                           |                              | LC50 = 2500 ppm. (Rat) 1h |
| Zinc oxide                  | LD50 > 5000 mg/kg ( Rat ) | LD50 > 2000 mg/kg, 24h (Rat) | LC50 > 5.7 mg/L, 4h (Rat) |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;  
**Respiratory** No data available  
**Skin** No data available

| Component                        | Test method                                        | Test species | Study result    |
|----------------------------------|----------------------------------------------------|--------------|-----------------|
| Zinc oxide<br>1314-13-2 ( 0.12 ) | in vivo OECD Test Guideline 406<br>Test method B.6 | guinea pig   | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                        | Test method                                                            | Test species         | Study result |
|----------------------------------|------------------------------------------------------------------------|----------------------|--------------|
| Zinc oxide<br>1314-13-2 ( 0.12 ) | in vitro<br>OECD Test Guideline 471<br>Bacterial Reverse Mutation Test | in vitro: Bacteria   | negative     |
|                                  | in vivo<br>OECD Test Guideline 474<br>Mammalian                        | in vivo<br>Mammalian | negative     |

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available  
**Target Organs** No information available.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## Section 12 - Ecological Information

### Ecotoxicity effects

| Component  | Freshwater Fish                             | Water Flea | Freshwater Algae | Microtox |
|------------|---------------------------------------------|------------|------------------|----------|
| Zinc oxide | LC50: = 1.55 mg/L, 96h static (Danio rerio) |            |                  |          |

**Persistence and Degradability** No information available

**Bioaccumulative Potential** No information available

| Component                   | log Pow | Bioconcentration factor (BCF) |
|-----------------------------|---------|-------------------------------|
| Nitric acid ...% [C ≤ 70 %] | -2.3    | No data available             |

**Mobility** No information available.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge.

## Section 14 - Transport Information

### IMDG/IMO

UN-No UN2031  
Proper Shipping Name NITRIC ACID  
Hazard Class 8  
Packing Group II

### ADG

UN-No UN2031  
Proper Shipping Name NITRIC ACID  
Hazard Class 8  
Packing Group II

| Component                                        | Hazchem Code    |
|--------------------------------------------------|-----------------|
| Nitric acid ...% [C ≤ 70 %]<br>7697-37-2 ( 6.5 ) | 2R<br>2P<br>2PE |

### IATA

UN-No UN2031  
Proper Shipping Name NITRIC ACID  
Hazard Class 8  
Packing Group II

Environmental hazards No hazards identified  
Special Precautions No special precautions required  
Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                               | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] - 7697-37-2 | Schedule 5 listed - except its salts and derivatives; in preparations except in preparations containing ≤0.5% of Nitric acid<br>Schedule 6 listed - except its salts and derivatives; except when included in Schedule 5, or in preparations containing ≤0.5% of Nitric acid                 |
| Zinc oxide - 1314-13-2                  | Schedule 4 listed - for human internal use except in preparations with a recommended daily dose of ≤25 mg of Zinc, or in preparations with a recommended daily dose of between 25-50 mg of Zinc when compliant with the requirements of the Required Advisory Statements for Medicine Labels |

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                               | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-----------------------------------------|-------------------------------------------------------------|------------------------|
| Water - 7732-18-5                       | Present                                                     | -                      |
| Nitric acid ...% [C ≤ 70 %] - 7697-37-2 | Present                                                     | -                      |
| Zinc oxide - 1314-13-2                  | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product contains one or more substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

| Component                               | Australian - Illicit Drug Precursors/Reagents Substance List | Chemicals of Security Concern                                                  |
|-----------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] - 7697-37-2 |                                                              | Listed in Appendix A<br>Precursors to homemade explosives - concentration ≥30% |

#### **Legend**

**Chemicals of Security Concern** - for further information see <http://www.chemicalsecurity.gov.au/securityconcerns>

**National pollutant inventory** Subject to reporting requirements



| Component                               | National pollutant inventory      |
|-----------------------------------------|-----------------------------------|
| Nitric acid ...% [C ≤ 70 %] - 7697-37-2 | 10 tonne/yr. Threshold category 1 |

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component                   | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Water                       | X    | X     | 231-791-2 | -      | X    | X   | -    | X     | X    |      | X     | KE-35400 |
| Nitric acid ...% [C ≤ 70 %] | X    | X     | 231-714-2 | -      | X    | X   | -    | X     | X    | X    | X     | KE-25911 |
| Zinc oxide                  | X    | X     | 215-222-5 | -      | X    | X   | -    | X     | X    | X    | X     | KE-35565 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component                               | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|-----------------------------------------|------------------------------------|------------------------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] - 7697-37-2 | Annex I - Y34                      | Y34 solid or solution                                                  |
| Zinc oxide - 1314-13-2                  | Annex I - Y23                      | Y23                                                                    |

| Component                   | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Water                       | 7732-18-5 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Nitric acid ...% [C ≤ 70 %] | 7697-37-2 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Zinc oxide                  | 1314-13-2 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component                   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Zinc oxide                  | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)            |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  |                                                                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### **Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

### **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

|                         |                 |
|-------------------------|-----------------|
| <b>Revision Date</b>    | 21-Nov-2022     |
| <b>Revision Summary</b> | Not applicable. |

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### **Disclaimer**

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## End of Safety Data Sheet